Project Report

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Project Title- Open Source Software Audit – Python

Introduction

Open Source Software (OSS) plays a crucial role in modern computing. It allows users to freely access, modify, and distribute software. This freedom promotes collaboration, transparency, and innovation.

Python is one of the most popular open-source programming languages in the world. It is widely used in various domains such as web development, artificial intelligence, data science, and automation.

This project focuses on analyzing Python as an open-source software. It includes both theoretical study and practical understanding through shell scripting in a Linux environment.

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2. Part A – Origin and Philosophy

A1: Origin of Python

Python was developed by **Guido van Rossum** and released in 1991.

During that time, programming languages were complex and difficult to use. Python was designed to simplify programming and improve code readability. The main goal was to make programming accessible to everyone.

Python emphasizes clean and readable code, which helps developers write efficient programs with fewer lines of code.

A2: License of Python

Python is distributed under the Python Software Foundation (PSF) License.

This license allows users to:

- Use the software freely
- Modify the source code
- Distribute the software

Python supports all four freedoms of open-source software:

1. Freedom to run the program
2. Freedom to study the source code
3. Freedom to modify the software
4. Freedom to distribute copies

A3: Ethics of Open Source

Open source software promotes collaboration and knowledge sharing. Developers from around the world contribute to improving software.

Advantages:

- Encourages innovation
- Builds community
- Provides transparency

Challenges:

- Companies may use open-source software for profit
- Lack of centralized control

Despite challenges, open source remains a powerful model for software development.

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3. Part B – Linux Footprint

Python is widely used in Linux operating systems.

- It is often pre-installed in Linux distributions
- It can be installed using package managers
- The executable is usually located in system directories

Python runs as a normal user, which enhances system security.

Linux provides a strong environment for Python development and execution. It allows users to manage packages, run scripts, and automate tasks efficiently.

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4. Part C – FOSS Ecosystem

Python is part of a large Free and Open Source Software (FOSS) ecosystem.

It supports various technologies:

- Web frameworks (Django, Flask)
- Data science tools (NumPy, Pandas)
- Machine learning libraries (TensorFlow, Scikit-learn)

Python depends on lower-level languages like C for performance.

It has a huge global community that contributes to its development. Developers collaborate through platforms like GitHub and forums.

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5. Part D – Open Source vs Proprietary Software

Feature	Open Source (Python)	Proprietary Software
Cost	Free	Paid
Source Code	Open	Closed
Flexibility	High	Limited

Customization	Allowed	Restricted
Community Support	Strong	Limited

Open-source software provides more flexibility and freedom compared to proprietary software.

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6. Practical Implementation

This project includes practical work using shell scripting in a Linux environment.

Five scripts were developed:

1. System Identity Report
2. Python Package Inspector
3. Disk and Permission Auditor
4. Log File Analyzer
5. Open Source Manifest Generator

These scripts demonstrate:

- System-level operations
- File handling
- Command usage
- Automation

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7. Learning Outcomes

This project helped in:

- Understanding open-source software concepts
- Learning about Python’s origin and features
- Gaining experience with Linux commands
- Developing shell scripting skills
- Improving logical thinking and problem-solving

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8. Conclusion

Python is one of the most powerful and widely used open-source programming languages. It provides simplicity, flexibility, and strong community support.

This project helped in understanding both theoretical and practical aspects of open-source software. It also highlighted the importance of Linux and shell scripting in real-world applications.

Overall, this project enhanced knowledge and provided valuable technical experience.

```
┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ ./script1.sh
═══════════════════════════
 Open Source Audit
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Student: Paras
Software: Linux
Distro: Kali GNU/Linux Rolling
Kernel: 6.12.38+kali-amd64
User: kali
Uptime: up 14 minutes
Date: Thu Mar 26 11:23:54 AM EDT 2026
License: GPL/MIT (based on OS)

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$
```

```
┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ sed -i 's/\r$//' script2.sh

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ chmod +x script2.sh

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ ./script2.sh
linux-image-6.12.38+kali-amd64 is installed
ii  linux-image-6.12.38+kali-amd64         6.12.38-1kali1                          amd64        Linux 6.12 for 64-bit PCs
No description available

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ sS
```

```
┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ sed -i 's/\r$//' script3.sh

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ chmod +x script3.sh

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ ./script3.sh
Directory Audit
/etc ⇒ drwxr-xr-x root root | Size: 15M
/var/log ⇒ drwxr-xr-x root root | Size: 47M
/home ⇒ drwxr-xr-x root root | Size: 2.4G
/usr/bin ⇒ drwxr-xr-x root root | Size: 704M
/tmp ⇒ drwxrwxrwt root root | Size: 8.0K
Checking /etc for configs ...
adduser.conf
alsa
alternatives
apache2
apparmor

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$
```

```
┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ sed -i 's/\r$//' script5.sh

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ chmod +x script5.sh

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$ ./script5.sh
Answer 3 questions
Tool you use: git
Freedom means: control
What you'd build: privacy focused version social platform
Saved to manifesto_kali.txt
On Thu Mar 26 11:27:52 AM EDT 2026, I believe open source means control. I use git daily and would build privacy focused version social platform and share it freely.

┌──(kali㉿kali)-[~/Desktop/Scripts/scripts]
└─$
```